

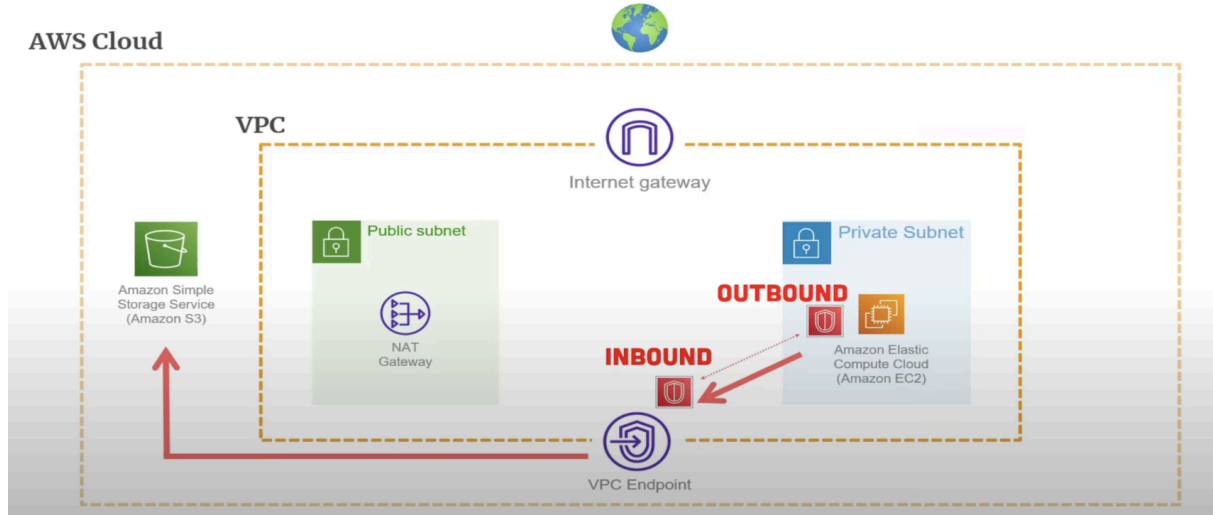
VPC Endpoint ile S3 Bucket'a erişim

Bir şirkette güvenlik gereği **private subnet içindeki EC2 instance**'ların internete çıkması yasak. Ancak bu EC2'lerin **S3 bucket'a (örn. log yedekleri için)** erişmesi gerekiyor.

- Normalde EC2 → S3 için internete çıkış (NAT Gateway/IGW) gerekir.
- Fakat burada **VPC Endpoint (Gateway Endpoint)** kullanarak EC2 → S3 iletişimini AWS'in private network'ü üzerinden sağlayacağız.

👉 Böylece:

- İnternete çıkış olmayacak, güvenlik artacak.
- Trafik AWS backbone üzerinden S3'e gidecek.
- Daha ucuz (NAT Gateway ücreti yok).





Lab Adımları

1 Ön Hazırlık – VPC & Subnet

1. AWS Console → VPC Dashboard → **Create VPC**

- VPC CIDR: 10.0.0.0/16
- Subnetler:
 - **Public Subnet:** 10.0.1.0/24
 - **Private Subnet:** 10.0.2.0/24
- Public Subnet'e IGW bağla (Bastion Host için kullanılabilir).
- Private Subnet → Route Table sadece Local entry ile olacak (0.0.0.0/0 yok).

VPC ve Subnet'ler

- AWS Console → VPC → “Create VPC”
 - **Vpc Settings:** VPC only
 - **Name tag:** my-VPC
 - **IPv4 CIDR block:** 10.0.0.0/16

Your VPCs > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

my-VPC

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info
Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name X	Q my-VPC X	Remove tag
Add tag		

You can add 49 more tags

Cancel [Preview code](#) [Create VPC](#)

Subnet oluşturma:

- Public Subnet

Subnets→Create Subnet

- **VPC ID:** my-VPC
- **Subnet Name:** public1
- **Availability Zone:** No preference
- **IPv4 subnet CIDR block:** 10.0.1.0/24

VPC

VPC ID
Create subnets in this VPC.

vpc-06b690e823ce74eab (my-VPC)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

public1
The name can be up to 255 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

▼ **Tags - optional**

Key	Value - optional
Name	public1

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

- Private Subnet

Subnets→Create Subnet

- **VPC ID:** my-VPC
- **Subnet Name:** private1
- **Availability Zone:** No preference
- **IPv4 subnet CIDR block:** 10.0.2.0/24

Subnet

VPC

VPC ID
Create subnets in this VPC.

vpc-06b690e823ce74eab (my-VPC)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

private1
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.240/24

▼ Tags - optional

Key	Value - optional	
Q Name	X	Q private1 X Remove

Add new tag
You can add 49 more tags.

Remove

Add new subnet

Cancel Create subnet

Internet Gateway (IGW)

VPC→Internet gateways→Create internet gateway

Internet gateway settings:

- Name: MY-IGW

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

MY-IGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

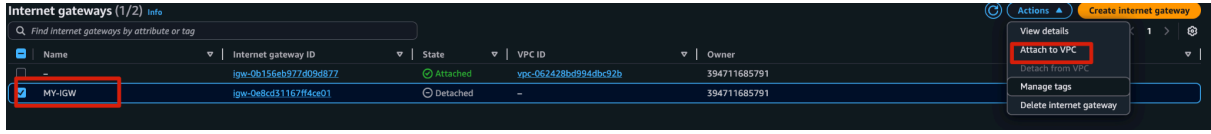
Key	Value - optional	
Q Name	X	Q MY-IGW X Remove

Add new tag
You can add 49 more tags.

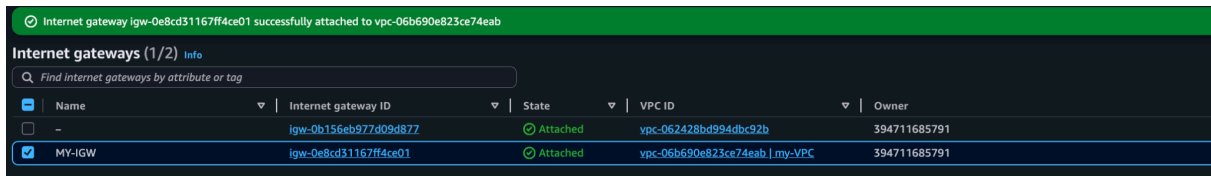
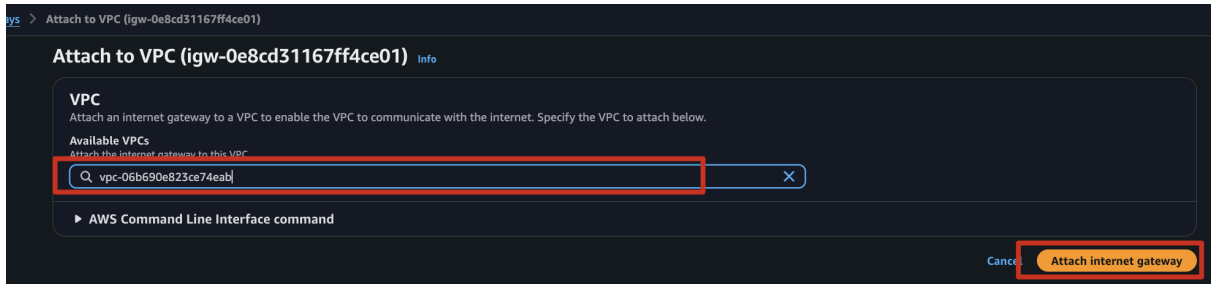
Cancel Create internet gateway

Internet gateway VPC'ye attach etme

internet gateway seç Actions dan Attach to VPC seç



VPC içerisinde **My-VPC** seç ve **Attach internet gateway** butonuna bas.

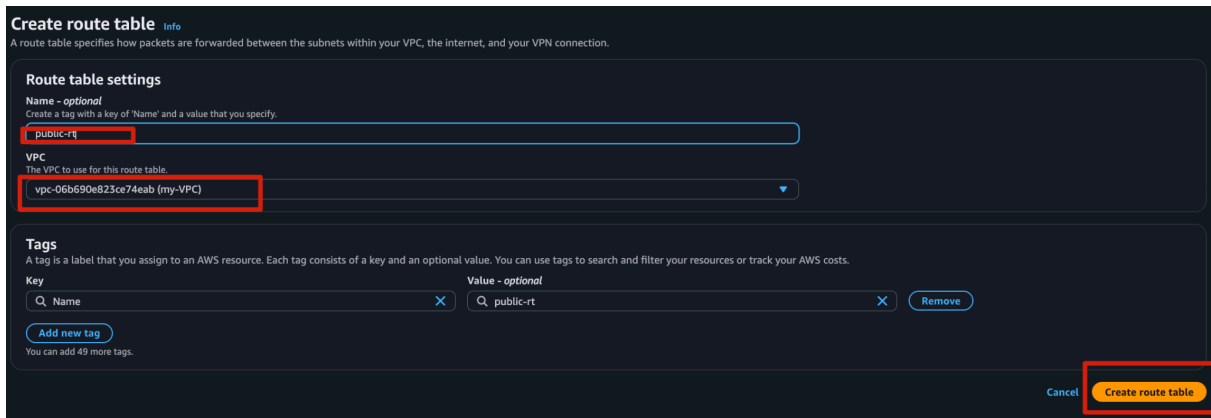


Route Tables

- Public Route Table:

VPC→Route tables→Create route table

- Name: **public-rt**
- VPC: **my-VPC**



Route Ekle

public-rt→Routes→Edit routes→Add route

- Destination: 0.0.0.0/0
- Target: internet Gateway(MY-IGW)

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway		No	CreateRouteTable

Buttons: Add route, Cancel, Preview, Save changes

Route Table'ı Public Subnet ile ilişkilendir

public-rt→Subnet associations→Edit subnet associations→Select Public-1

- Available subnets: public-1

Edit subnet associations

Change which subnets are associated with this route table.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
private-1	subnet-0f2f49cf31af6d9d1	10.0.2.0/24	-
public-1	subnet-0e6b60ab2bf380fa	10.0.1.0/24	-

Selected subnets: subnet-0e6b60ab2bf380fa / public-1

Buttons: Cancel, Save associations

VPC→Route tables→Create route table

- Name: private-rt
- VPC: my-VPC

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key: Name, Value: private-rt

Buttons: Add new tag, Cancel, Create route table

Route Table'ı Private Subnet ile ilişkilendir

private-rt→Subnet associations→Edit subnet associations→Select private-1

- Available subnets: **private-1**

b-Ofcc847a3e8a4b406 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
☒	private-1	subnet-0f2f49cf31af6d9d1	10.0.2.0/24	-
☐	public-1	subnet-0e6e6b0ab2bf380fa	10.0.1.0/24	-

Selected subnets

subnet-0f2f49cf31af6d9d1 / private-1 X

Cancel Save associations

2 EC2 Instance Kurulumu

1. **Public Subnet**'te bir Bastion EC2 instance oluřtur:

EC2→Instances→Launch an instance

- **Name and tags:** Bastion
- **AMI**→ Amazon Linux 2023, t3.micro

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Bastion

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

Debian

debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-052064a798f08f0d3 (64-bit (x86), uefi-preferred) / ami-089f6a79b0e02648a (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.9.20250929.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86)	uefi-preferred	ami-052064a798f08f0d3	2025-09-25	ec2-user	Verified provider

▼ Instance type Info | Get advice

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

- **Key Pair (login):**
 - **Create new key pair** → **Key pair name:** `ec2_ssh_key`
 - **Type:** RSA, **File format:** `.pem` → **Create ve seç.**

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

ec2_ssh_key ▼

↻ Create new key pair

Create key pair ✕

Key pair name
Key pairs allow you to connect to your instance securely.

ec2_ssh_key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

- **Network settings** → **Edit:**
 - **VPC:** `my-VPC`
 - **Subnet:** `public-1`
 - **Auto-assign public IP:** `Enable`
 - **Firewall (SG):** Select security group
 - **Security group name:** `bastion-sg`
 - **Description:** `bastion-sg`
 - **Inbound rules:**
 - **Type:** `SSH`
 - **Protocol:** `TCP`
 - **Port range:** `22`
 - **Source:** `0.0.0.0/0`

▼ Network settings Info

VPC - required Info

vpc-06b690e823ce74eab (my-VPC)
10.0.0.0/16

Subnet Info

subnet-0e6e6b0ab2bf380fa
VPC: vpc-06b690e823ce74eab Owner: 394711685791 Availability Zone: us-east-1e (use1-az3)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24 public-1

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

bastion-sg

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@!+=&()*!\$*

Description - required Info

bastion-sg

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type	Protocol	Port range	Source type	Source	Description - optional
ssh	TCP	22	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Launch instance butonuna tıklayınız

Instances (1) Info

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	Bastion	i-0222278277ec1022b	Running	t2.micro	Initializing	View alarms +	us-east-1e	-	54.209.65.163	-

2. Private Subnet'te bir EC2 instance oluşturun:

EC2→Instances→Launch an instance

- **Name and tags:** `ec2-s3`
- **AMI**→ `Amazon Linux 2023`, `t2.micro`
- **Key pair name:** `ec2_ssh_key`

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

`ec2-s3` [Add additional tags](#)

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start



Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linux



Debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-052064a798f08f0d3 (64-bit (x86), uefi-preferred) / ami-089f6a79b0e02648a (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.9.20250929.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-052064a798f08f0d3	2025-09-25	ec2-user

Verified provider

Instance type

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

`ec2_ssh_key` [Create new key pair](#)

- Network settings → Edit:
 - VPC: my-VPC
 - Subnet: private-1
 - Auto-assign public IP: Disable
 - Firewall (SG): Select security group
 - Security group name: ec2-s3-sg
 - Description: ec2-s3-sg
 - Inbound rules:
 - Type: SSH
 - Protocol: TCP
 - Port range: 22
 - Source: bastion-sg

▼ Network settings Info

VPC - required Info

vpc-06b690e823ce74eab (my-VPC)
10.0.0.0/16

Subnet Info

subnet-0f2f49cf31af6d9d1 private-1
VPC: vpc-06b690e823ce74eab Owner: 394711685791 Availability Zone: us-east-1e (use1-az3)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

ec2-s3-sg

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .-:/()#,@!+=&:~*`

Description - required Info

ec2-s3-sg

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-0a75c80668a3ad5ab) Remove

Type	Protocol	Port range	Source type	Source	Description - optional
ssh	TCP	22	Custom	sg-0a75c80668a3ad5ab	e.g. SSH for admin desktop

Launch instance butonuna bas

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
ec2-s3	i-08d69df94305a336a	Running	t2.micro	Initializing	View alarms +	us-east-1e	-	-	-	-
Bastion	i-0222278277ec1022b	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1e	-	54.209.65.163	-	-

3. IAM servisinde IAM Role oluřtur

- IAM Role: S3 eriřim yetkisi olan bir role ata (**AmazonS3ReadOnlyAccess**)

IAM→Role→Create Role

AWS Service seę→EC2 seę ve **next** butonuna bas.

The screenshot shows the 'Create role' page in the AWS IAM console. The 'Trusted entity type' section has 'AWS service' selected. The 'Use case' section has 'EC2' selected. The 'Next' button is highlighted in red.

AmazonS3ReadOnlyAccess permission seę ve **next** butonuna bas

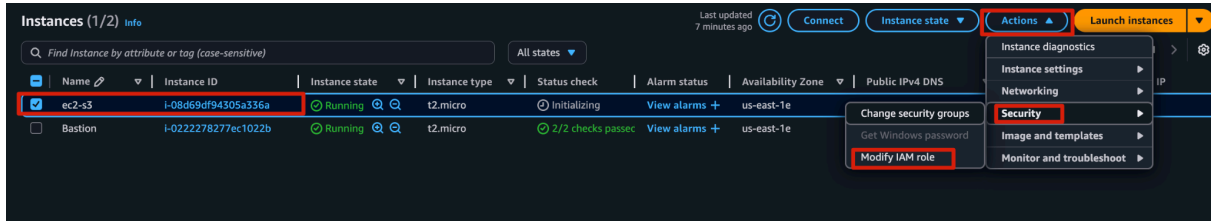
The screenshot shows the 'Add permissions' page in the AWS IAM console. The 'Permissions policies' section shows 'AmazonS3ReadOnlyAccess' selected. The 'Next' button is highlighted in red.

Dięer ayarları default bırak ve **create role** butonuna bas

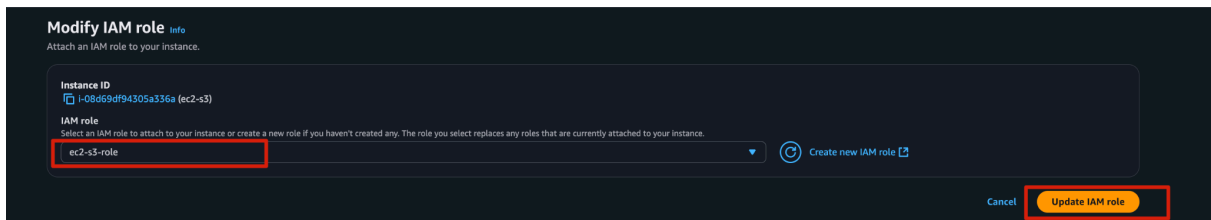
The screenshot shows the 'Name, review, and create' page in the AWS IAM console. The 'Role name' is 'EC2-3-role'. The 'Trust policy' section shows the default trust policy. The 'Create role' button is highlighted in red.

4. EC2 suncuya Role ata

- **Actions→Security→Modify IAM role**



- **Ec2-s3-role rseç Update IAM role butonuna bas**

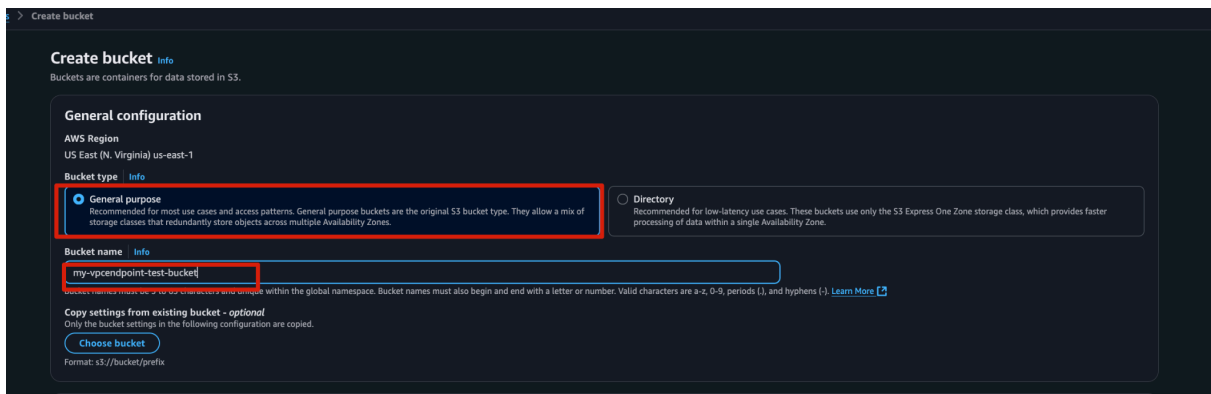


3 S3 Bucket Hazırlığı

1. S3 Bucket Oluşturma

Amazon S3→Create Bucket

Name : **my-vcendpoint-test-bucket**



Object Ownership

info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block all public access

Turning on Block all public access turns on all four settings below. Each of the following settings are independent of one another.

☒ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

info

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☒ Disable

☐ Enable

Tags - optional (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

No tags associated with this bucket.

Add new tag

You can add up to 50 tags.

Default encryption

info

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

info

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket

S3 bucket ismi verdikten sonra herşeyi default ayarlarında bırak ve **Create bucket** butonuna basarak s3 bucket oluştur.

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1)

info

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

⚙

Name	AWS Region	Creation date
☐ my-vcendpoint-test-bucket	US East (N. Virginia) us-east-1	October 3, 2025, 21:11:30 (UTC+03:00)

Account snapshot

info

Updated daily

View dashboard

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new

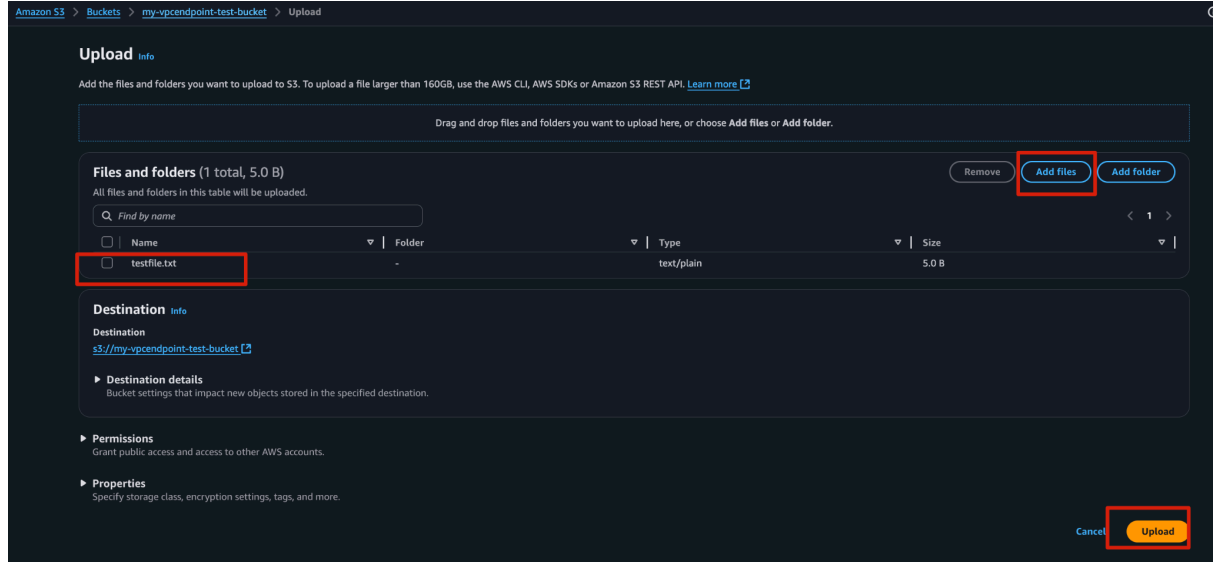
info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

2. S3'e Dosya Yükleme

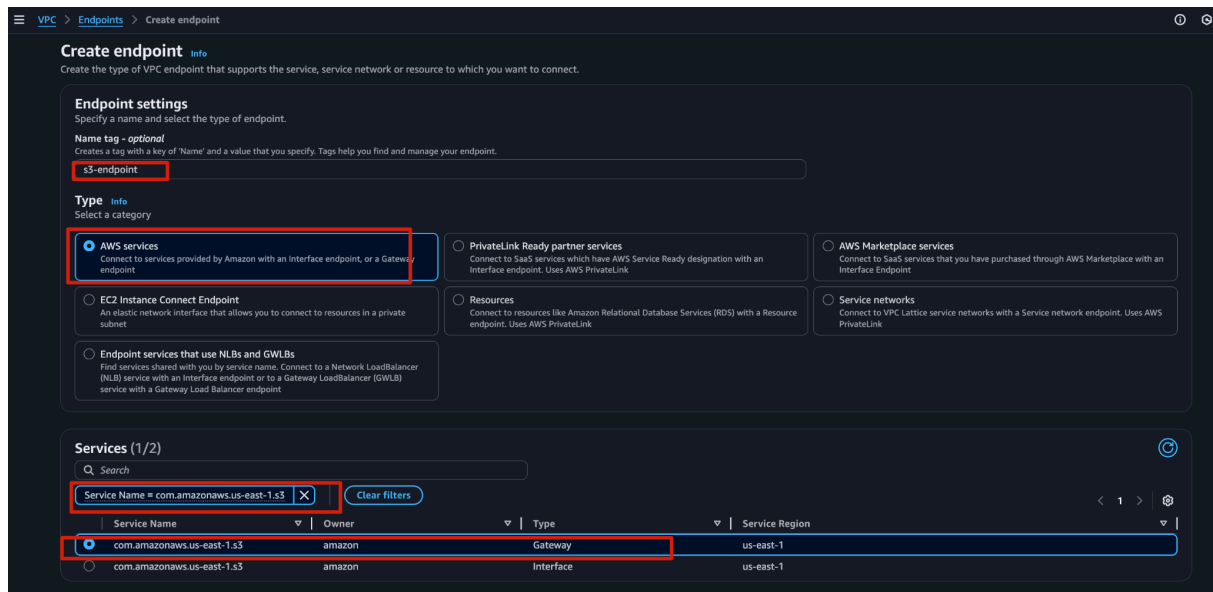
(my-vpcendpoint-test-bucket) Bucket'ı seçerek upload butonuna tıklave s3 bucket içine testfile.txt adında dosyayı ekle.



4 VPC Endpoint (Gateway) Kurulumu

1. VPC→PrivateLink and Lattice→ Endpoints → Create Endpoint

- **Name:** s3-endpoint
- **Service category:** AWS services
- **Service Name:** com.amazonaws.us-east-1.s3
- **Type:** Gateway



- **VPC:** my-VPC
- **Route Table:** Private Subnet'in Route Table'ını seç.
- Policy: Full Access (test için).

Network settings
Select the VPC in which to create the endpoint.

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.
vpc-06b690e823ce74eab (my-VPC)

Route tables (1/3) Info

Name	Route Table ID	Main	Associated Id
-	rtb-09cab56ffcf59a9ed	Yes	-
public-RT	rtb-0bd6536519cae4b21 (public-RT)	No	subnet-0e6e6b0ab2bf380fa (public-1)
☒ private-RT	rtb-0fcc847a3eba4b406 (private-RT)	No	subnet-0f2f49cf31afd9d1 (private-1)

When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.

Policy Info
VPC endpoint policy controls access to the service.

☒ Full access
Allow access by any user or service within the VPC using credentials from any Amazon Web Services accounts to any resources in this Amazon Web Services service. All policies — IAM user policies, VPC endpoint policies, and Amazon Web Services service-specific policies (e.g. Amazon S3 bucket policies, any S3 ACL policies) — must grant the necessary permissions for access to succeed.

☐ Custom
Use the policy creation tool to generate a policy, then paste the generated policy below.

Create endpoint butonuna tıkla

Successfully created VPC endpoint
vpce-08db146130f33c7a7

Endpoints (1/1) Info

Find endpoints by attribute or tag

VPC endpoint ID : vpce-08db146130f33c7a7

Name	VPC endpoint ID	Endpoint type	Status	Service name	Service network
s3-endpoint	vpce-08db146130f33c7a7	Gateway	Available	com.amazonaws.us-east-1.s3	-

📌 Bu işlem sonunda Route Table'a otomatik şu entry eklenir:

pl-xxxx (Prefix List for S3) → Target = VPC Endpoint

tb-0fcc847a3eba4b406 / private-RT

Destination	Target	Status	Propagated	Route Origin
pl-63a5400a	vpce-08db146130f33c7a7	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

5 Test – EC2’den S3 Erişimi

1. Bastion Host üzerinden Private EC2’ye SSH ile bağlan.

```
ssh -i ec2_ssh_key.pem -o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@<Bastion_PublicIP>' ec2-user@<ec2-s3-privateIP>
```

```
chmod 400 ec2_ssh_key.pem
ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@54.209.65.163' \
ec2-user@10.0.2.14
```

```
(base) Downloads ➤ chmod 400 ec2_ssh_key.pem
(base) Downloads ➤ ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@54.209.65.163' \
ec2-user@10.0.2.14
The authenticity of host '54.209.65.163 (54.209.65.163)' can't be established.
ED25519 key fingerprint is SHA256:zmbq86Xg5lKYXQqytinRh2o7zc0qqKVMeJQ9oSV2SKo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.209.65.163' (ED25519) to the list of known hosts.
The authenticity of host '10.0.2.14 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:APzYr96jJRpankk574qNC+qvzbT0drnARpuz2nxSaPg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.14' (ED25519) to the list of known hosts.

#_
~\_ #####_ Amazon Linux 2023
~~ \_#####\
~~ \_###l
~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~
~~~.._/_/_/
~~/_/_/_/
~~/_m/'
[ec2-user@ip-10-0-2-14 ~]$
```

AWS CLI ile test et:

```
aws s3 ls s3://my-vpcendpoint-test-bucket
aws s3 cp s3://my-vpcendpoint-test-bucket/testfile.txt .
```

```
[ec2-user@ip-10-0-2-14 ~]$ aws s3 ls s3://my-vpcendpoint-test-bucket
2025-10-03 19:15:20          5 testfile.txt
[ec2-user@ip-10-0-2-14 ~]$ aws s3 cp s3://my-vpcendpoint-test-bucket/testfile.txt .
download: s3://my-vpcendpoint-test-bucket/testfile.txt to ./testfile.txt
[ec2-user@ip-10-0-2-14 ~]$ ls -l
total 4
-rw-r--r--. 1 ec2-user ec2-user 5 Oct  3 19:15 testfile.txt
[ec2-user@ip-10-0-2-14 ~]$
```

2. Eğer internet yok ama dosya iniyorsa → VPC Endpoint başarılı 🎉

```
[ec2-user@ip-10-0-2-14 ~]$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
```



Notlar

- **IAM Role olmadan** EC2, S3'e erişemez → mutlaka Role/Policy ekle.
- Bu senaryoda **NAT Gateway'e ihtiyaç yok** → maliyet düşer.